

SECTION A [25 marks]

This section consists of 3 questions. Answer all the questions given.

1. (a) Given $f(x) = \frac{x^2 - 1}{x - 1}$, find $\lim_{x \rightarrow 1} f(x) - \lim_{x \rightarrow 0} f(x)$. [4 marks]

(b) Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - 1}}{3x + 5}$ [5 marks]

2. (a) Given that $y = x \ln x - 3x$, find the value of x and y when $\frac{dy}{dx} = 0$. [5 marks]

(b) Given $y = \alpha e^{2x} + \beta e^{-2x}$ where α and β are constants. Show that $\frac{d^2 y}{dx^2} = 4y$. [6 marks]

3. Given $f(x) = x^2 - \frac{2}{x}$, where $x \neq 0$. Find the coordinate of stationary point and determine whether it is relative maximum or relative minimum. [5 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer all questions.

1. Given that $z_1 = 2 + 3i$, $z_2 = 3 + 2i$ and $z_3 = a + bi$ are three complex numbers, where a and b are real.

(a) Find $z_1 + z_2$. Hence, determine the modulus of $z_1 + z_2$.

(b) If $z = \frac{z_1 z_3}{z_2}$, find z in terms of a and b . Giving your answer in the form $x + yi$, where $x, y \in \mathbb{R}$.

(c) From part (b), given that $z = \frac{17}{13} - \frac{7}{13}i$,

- (i) find the values of a and b .
- (ii) find $\arg z$, giving your answer in radians to 3 decimal places.
- (iii) Hence, express z in polar form.

[9 Marks]

2. (a) Solve the equation $\log_5 5x + \log_5 5 = \log_5 64$.
- (b) Determine the value of x for $3 - x \leq x(x - 3) < 10 - 3x$. Express your answer in the interval form.

[11 Marks]

3. The function f is defined by $f(x) = \frac{x+2}{x-2}$, $x \neq k$ and $g(x) = \frac{2x+2}{x-1}$
- (a) State the value of k .
 - (b) Determine whether f is a one-to-one function algebraically.
 - (c) Find $f \circ g$ and hence state the relationship between f and g .

[8 marks]

4. The functions f and g are defined by $f(x) = \ln(3x - 1)$, $g(x) = e^x$
- (a) Determine the inverse function of f .
 - (b) State the domain and range of f^{-1} .
 - (c) Sketch the graphs of f and f^{-1} on the same axes.
 - (d) Find $g \circ f$.

[10 marks]

5. Given that

$$f(x) = \begin{cases} \frac{(2-x)^2 - p}{x}, & x < q \\ r(x+6), & q \leq x < 2 \\ x^3 - p, & x \geq 2 \end{cases}$$

find the values of p , q and r such that $f(2) = p$ and $f(x)$ is continuous everywhere.

[11 marks]

6. (a) A curve has the equation $y^2 + 6x = 3xy + y$. Find the coordinates of the points on the curve and its gradient when $x = 2$. [8 marks]
- (b) The parametric equations of a curve are given by $x = \frac{1}{3} \sin^3 2\theta$ and $y = 2 \cos^3 2\theta$. Find $\frac{dy}{dx}$ in terms of θ . Hence, find the value of $\frac{dy}{dx}$ when $\theta = \frac{\pi}{6}$. [8 marks]
7. Water is leaking from the bottom of a conical tank with radius 1.5 meter and height 2 meter at a rate of 0.25 cubic meter per minute. The tank was initially full. If the height of water is 1 meter then, find the rate of change of
- (a) the water level.
- (b) the radius of the water surface. [9 marks]

ANSWER

Section A

1. (a) 1 (b) $\frac{1}{3}$
2. (a) $x = e^2$ $y = -e^2$ (b) shown
3. Stationary point $(-1, 3)$, relative minimum.

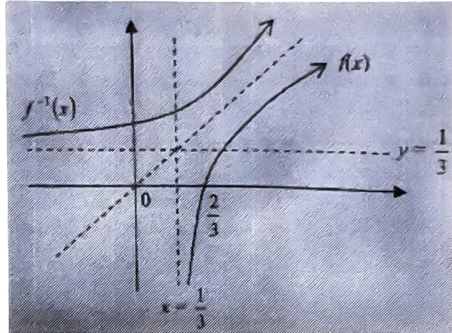
Section B

1. (a) $5 + 5i$, $5\sqrt{2}$ (b) $\frac{12a-5b}{13} + \frac{(5a+12b)}{13}i$
- (c) (i) $a=1, b=-1$ (ii) $\arg z = -0.391 \text{ rad}$
- (iii) $z = \sqrt{2}(\cos(-0.391 \text{ rad}) + i \sin(-0.391 \text{ rad}))$
2. (a) $x = 5$ (b) $(-\sqrt{10}, -1] \cup [3, \sqrt{10})$
3. (a) $k = 2$
- (c) $f[g(x)] = x, \therefore f$ and g are inverse each other

4. (a) $f^{-1}(x) = \frac{e^x + 1}{3}$

(b) $D_{f^{-1}} = R_f = (-\infty, \infty), R_{f^{-1}} = D_f = (\frac{1}{3}, \infty)$

(c)



(d) $3x - 1$

5. $p = 4, r = \frac{1}{2}, q = 0$

6. (a) $(2, 4)$ and $(2, 3)$; gradient = $6, -3$

(b) $\frac{dy}{dx} = -6 \cot 2\theta ; -2\sqrt{3}$

7. (a) -0.1415 m/min

(b) -0.1061 m/min